Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the f1 score by dataset.

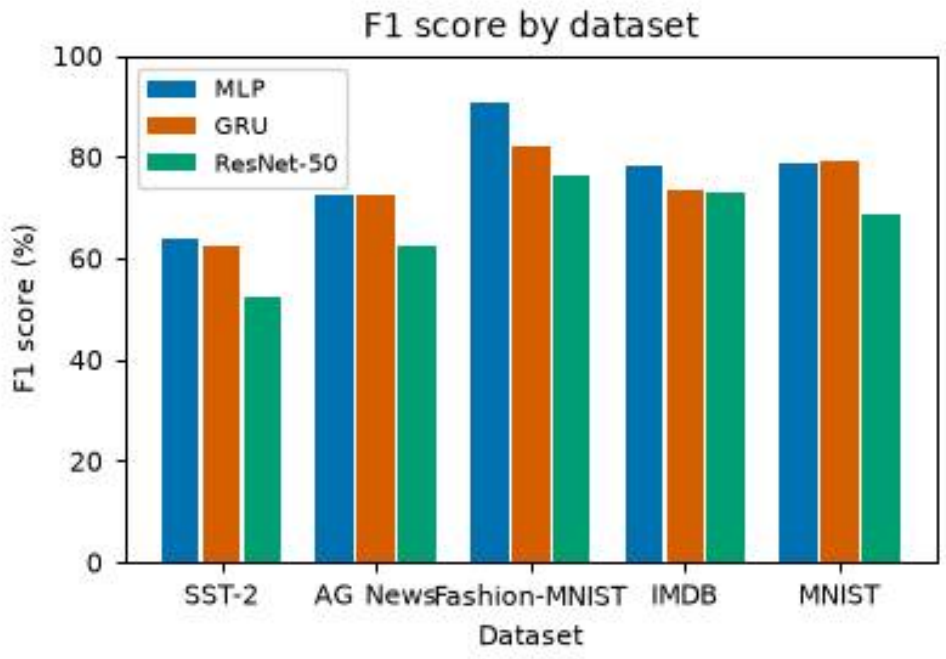


Figure 1: F1 score by dataset for the models in this experiment.

GRU achieves 62.6% F1 score on SST-2 (Figure 1). GRU reaches 76.9% F1 score on Fashion-MNIST. MLP reaches 85.7% F1 score on Fashion-MNIST.

Figure 2 shows the f1 score during training.

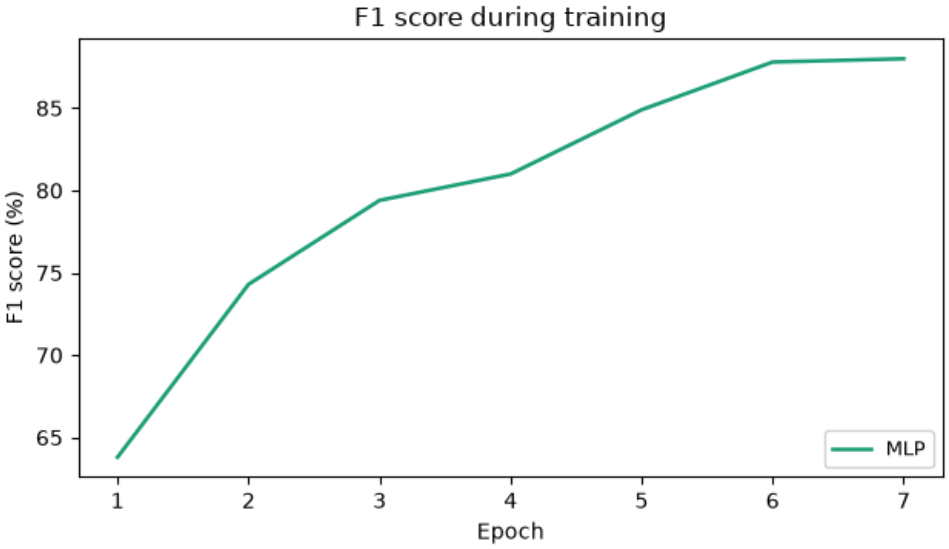


Figure 2: F1 score during training for the models in this experiment.

As shown in Figure 2, MLP reaches 61.0% F1 score at epoch 2. MLP reaches 81.0% F1 score at epoch 4 (Figure 2). MLP reaches 63.8% F1 score at epoch 2.

3 Discussion

On SVHN, Logistic Regression obtains an accuracy of 71.1%. We note that the training was stable for all models.